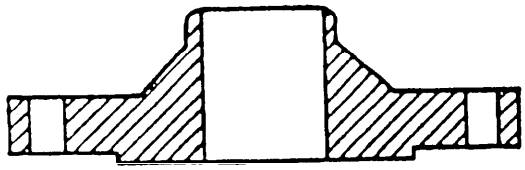
MILITARY SPECIFICATION SHEET

FLANGES AND FLANGED FITTINGS, PIPE, STEEL, CLASSES 150, 300, 400, 600, 900, 1500, AND 2500; WELDING NECK FLANGES

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the welding neck flanges described herein shall consist of this document and the latest issue of specification MIL-F-18180.



ANSI B16.5



CONVOLUTED

FIGURE 1. Welding neck flanges.

1. Military specification sheet part number. Welding neck flanges covered by this specification sheet shall be designated in the following form (see 6.2(b) of MIL-F-18180).

M18180/1	XXX	X	XX	X	X
Military Specification Sheet Code Number					
	Pipe Size and Pressure-Temperature Rating Class Code Number (see para 2)				
		Flange Facing Code Letter (see para 3)			
			ANSI B16.5 Materials Group No. Code Number (see para 4)		
				Pipe Schedule Code Letter (see para 5)	
					Reduce Size Code Letter (for Reducing Welding Neck Flanges Only (see para 6)

FSC 4730

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2. Pipe size and pressure-temperature rating. The nominal pipe size and pressure-temperature rating class of welding neck flanges are designated by three digits (see table I).

TABLE I. Code number to size and class.

Nominal Pipe Size (inch)	Pressure-Temperature Ratings						
	Class 150	Class 300	Class 400	Class 600	Class 900	Class 1500	Class 2500
1/2	001	021	Code No. for same size in class 600	052	Code No. for same size in class 600	084	103
3/4	002	022		053		085	104
1	003	023		054		086	105
1-1/4	004	024		055		087	106
1-1/2	005	025		056		088	107
2	006	026		057		089	108
2-1/2	007	027		058		090	109
3	008	028		059	072	091	110
3-1/2	009	029		060	---	---	---
4	010	030	041	061	073	092	111
5	011	031	042	062	074	093	112
6	012	032	043	063	075	094	113
8	013	033	044	064	076	095	114
10	014	034	045	065	077	096	115
12	015	035	046	066	078	097	116
14	016	036	047	067	079	098	---
16	017	037	048	068	080	099	---
18	018	038	049	069	081	100	---
20	019	039	050	070	082	101	---
24	020	040	051	071	083	102	---

3. Flange facing. The facing of welding neck flanges is designated by a single letter (see table II).

TABLE II. Code letter to flange facing.

Code Letter	Flange Facing
A	Raised
B	Flat
C	Ring-Joint
D	Large Male
E	Large Female
F	Small Male
G	Small Female
H	Large Tongue
J	Small Tongue
K	Large Groove
M	Small Groove

4. ANSI B16.5 material group number. The ANSI B16.5 Material Group Number for welding neck flanges is designed by two digits (see table III).

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TABLE III. Code number to material group number.

Code Number	ANSI B16.5 Material Group No.	ANSI B16.5 Acceptable Material (ASTM Standards for)
01	Group 1.1	A105, A216 Grade WCB, A350 Grade LF2, A515 Grade 70, A516 Grade 70, or A537 Class 1.
02	Group 1.2	A203 Class B or E, A216 Grade WCC, A350 Grade LF3 or A352 Grade LC2 or LC3.
04	Group 1.4	A350 Grade LF1, A515 Grade 60, or A516 Grade 60
05	Group 1.5	A182 Grade F1, A204 Grade A or B, A217 Grade WC1, or A352 Grade LC1
07	Group 1.7	A182 Grade F2, A204 Grade C, or A217 Grade WC 4 or WC 5.
09	Group 1.9	A182 Grade F11 or F12, A217 Grade WC6, or A387 Grade 11.
10	Group 1.10	A182 Grade F22, A217 Grade WC9, or A387 Grade 22.
13	Group 1.13	A182 Grade F5a, or A217 Grade C5.
14	Group 1.14	A182 Grade F9, or A217 Grade C12.
21	Group 2.1	A182 Grade F304 or F304H, A240 Type 304, or A351 Grade CF3 or CF8.
22	Group 2.2	A182 Grade F316 or F316H, A240 Type 316 or 317, or A351 Grade CF3M or CF8M.
23	Group 2.3	A182 Grade F304L or F316L, or A240 Type 304L or 316L.
24	Group 2.4	A182 Grade F321 or F321H, or A240 Type 321 or 321H.
25	Group 2.5	A182 Grade F347, F347H, F348 or F348H, A240 Type 347, 347H, 348 or 348H or A351 Grade CF8C.
26	Group 2.6	A240 Type 309S, or A351 Grade CH8 or CH20.
27	Group 2.7	A182 Grade F310, A240 Type 310S, or A351 Grade CK20.

5. Pipe schedule. The pipe schedule which is used to specify the waterway bore diameter (inside diameter of pipe for schedule specified) of welding neck flanges is designated by a single letter (see table IV).

NOTE: The waterway bore diameter for Class 150 (1/2 inch through 12 inch) and class 300 (1/2 inch through 12 inch) welding neck flanges is given in ANSI B16.5 and is the inside diameter of standard wall pipe.

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TABLE IV. Code letter to pipe schedule.

Code Letter	Pipe Schedule (for pipe sizes given)
A	Schedule 40 and Standard Wall (1/2 inch thru 10 inch)
B	Schedule 40 (12 inch and larger)
C	Standard Wall (12 inch and larger)
D	Schedule 80 and Extra Strong Wall (1/2 inch thru 8 inch)
E	Schedule 80 (10 inch and larger)
F	Extra Strong Wall (10 inch and larger)
G	Schedule 160
H	Double Extra Strong Wall (1/2 inch thru 8 inch)
J	Schedule 10 (14 inch and larger)
K	Schedule 20 (8 inch and larger)
M	Schedule 30 (8 inch and larger)
N	Schedule 60 (8 inch and larger)
R	Schedule 100 (8 inch and larger)
S	Schedule 120 (4 inch and larger)
T	Schedule 140 (8 inch and larger)
W	Schedule 5S (stainless steel pipe only)
X	Schedule 10S (stainless steel pipe only)

6. Reduced size. Reducing welding neck flanges and the size to which the reduction is being made are identified by a single letter (see table V).

TABLE V. Code letter to reduced nominal pipe size.

Code Letter	Reduced size (nominal pipe size-inch)
A	1/2
B	3/4
C	1
D	1-1/4
E	1-1/2
F	2
G	2-1/2
H	3
J	3-1/2
K	4
M	5
N	6
P	8
R	10
S	12
T	14
U	16
W	18
X	20

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7. Military specification part number definition. The military specification sheet part number is a definitive part number which is formed by combining the military specification sheet code number (M18180/1 for MIL-F-18180/1) the pipe size and pressure-temperature rating class code number, the flange facing code letter, the ANSI B16.5 materials group number code number, the pipe schedule code letter and, if applicable, the reduced size code letter as shown in paragraph 1.

Custodians:

Army - ME
Navy - YD

Review Activity:

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User Activities:

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Preparing Activity:

Navy - YD

(Project 4730-0704)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-18180//A	FLANGES AND FLANGED FITTINGS, PIPE, STEEL, CLASSES 150, 300, 400, 600, 900, 1500, AND 2500

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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